

1.

Create table dept1 as below with following columns: DEPTNO INTEGER PK,DNAME VARCHAR2(30) NOT NULL,LOC VARCHAR2(30) NOT NULL

```
SQL> create table dept1(deptno integer primary key,dname varchar2(30) not null, loc varchar2(30) not null);
Table created.
```

2.

Create table emp1 with the following columns EMPNO INTEGER PK,ENAME VARCHAR2(20) NOT NULL,SAL NUMBER(10,2) check(sal>5000),mgr number FK(EMPNO),DEPTNO INTEGER FK(DEPTNO)

```
SQL> create table emp1(empno integer primary key,ename varchar2(20) not null,sal number(10,2) check (sal>5000),mgr number,deptno integer, foreign key(deptno) references dept1(deptno));
Table created.
```

3.

Create tables Dept11,Emp11 from Dept1,Emp1.

```
SQL> desc dept11;
Name                                     Null?      Type
-----
DEPTNO                                              NUMBER(38)
DNAME                                     NOT NULL   VARCHAR2(30)
LOC                                       NOT NULL   VARCHAR2(30)

SQL> create table emp11 as select * from emp1;
Table created.

SQL> desc emp11;
Name                                     Null?      Type
-----
EMPNO                                              NUMBER(38)
ENAME                                     NOT NULL   VARCHAR2(20)
SAL                                                NUMBER(10,2)
MGR                                                NUMBER
DEPTNO                                              NUMBER(38)

SQL> alter table emp1 add Address varchar2(30);
Table altered.
```

4.

Add a new column Address-Varchar2(30) in the Emp1 table.

```
SQL> alter table emp1 add Address varchar2(30);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

5.

Rename SAL column in Emp1 table to SALARY.

```
SQL> alter table emp1 rename column sal to salary;
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

6.

Rename the FK constraint names and verify the names for the EMP1.

```
SQL> alter table emp1  
2 rename constraint SYS_C008260 to FK_DEPT;
```

Table altered.

```
SQL> select constraint_name, constraint_type  
2 from user_constraints  
3 where table_name='EMP1';
```

CONSTRAINT_NAME
C
FK_DEPT
R
SYS_C008257
C
SYS_C008258
C
SYS_C008259
P

7.

Modify ENAME Column by increasing the length of the column to 40.

```
SQL> alter table emp1 modify ename varchar2(40);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(40)
SALARY		NUMBER(10,2)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

8.

Drop the NOT NULL constraint for ENAME.

```
SQL> ALTER TABLE emp1 MODIFY ename varchar2(40) NULL;
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME		VARCHAR2(40)
SALARY		NUMBER(10,2)
MGR		NUMBER

9.

Create comment on Dept1 table as 'Depts of WIPRO'

```
SQL> comment on table dept1 is 'Depts of WIPRO';  
Comment created.
```

10.

Create comment on Deptno of Dept1 as 'Deptno of WIPRO'

```
SQL> comment on column dept1.deptno is 'deptno of WIPRO';  
Comment created.
```

11.

Create a comment on Emp1 table as 'Employees of WIPRO'

```
SQL> comment on table emp1 is 'Employees of WIPRO';
Comment created.
```

12.

Create a comment on Empno of Emp1 table as 'Empno of WIPRO'

```
SQL> comment on column emp11.empno is 'empno of WIPRO';
Comment created.
```

13.

Remove all the comment on the tables and columns.

```
SQL> COMMENT ON table emp1 IS ' ';
Comment created.
SQL> COMMENT ON COLUMN emp1.empno IS ' ';
Comment created.
SQL> COMMENT ON table dept1 IS ' ';
Comment created.
SQL> COMMENT ON COLUMN dept1.deptno IS ' ';
Comment created.
```

14.

Set sal,ename columns of Emp1 table as unused.

```
SQL> ALTER TABLE emp1 SET UNUSED (salary,ename);
```

Table altered.

```
SQL> desc emp1;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
MGR		NUMBER
DEPTNO		NUMBER(38)
ADDRESS		VARCHAR2(30)

15.

Drop the unused columns.

```
SQL> alter table emp1 drop unused columns;
Table altered.
```

16.

Drop Emp1,Dept1 Tables.

```
SQL> drop table dept1;  
Table dropped.
```

```
SQL> drop table dept1;  
Table dropped.
```

17.

Create Table Emp1 from EMP table by copying data also. Check your data.

```
SQL> create table emp1 as select * from emp;  
Table created.  
SQL> select * from emp1;
```

EMPNO	ENAME	JOB	MGR	HIREDATE
SAL	COMM	DEPTNO		
7839 5000	KING	PRESIDENT 10		17-NOV-81
7788 2850	BLAKE	MANAGER 30	7839	01-MAY-81
7799 2855	JOHNSON	MANAGER 30	7342	01-JUN-87

Rename Emp1 to EMP_Test.

18.

```
SQL> alter table emp1 rename to emp_test;  
Table altered.  
SQL> select * from emp_test;
```

EMPNO	ENAME	JOB	MGR	HIREDATE
SAL	COMM	DEPTNO		
7839 5000	KING	PRESIDENT 10		17-NOV-81
7788 2850	BLAKE	MANAGER 30	7839	01-MAY-81
7799 2855	JOHNSON	MANAGER 30	7342	01-JUN-87

19.

Truncate Emp_Test table and confirm your delete.

```
SQL> truncate table emp_test;

Table truncated.

SQL> select * from emp_test;

no rows selected
```

20.

Create Emp2 from Emp by only copying Empno,Ename,sal columns by copying Data.

```
SQL> create table emp2 as select empno,ename,sal from emp;

Table created.

SQL> desc emp2;
  Name                               Null?    Type
-----
EMPNO                                NUMBER(4)
ENAME                                VARCHAR2(20)
SAL                                  NUMBER(7,2)
```

21.

Drop Emp2 Table.

```
SQL> drop table emp2;

Table dropped.

SQL> desc emp2;
ERROR:
ORA-04043: object emp2 does not exist
```

22.

Create Emp2 table again from EMP table without copying data.

```
SQL> create table emp2 as select empno,ename,sal from emp;

Table created.

SQL> desc emp2;
  Name                               Null?    Type
-----
EMPNO                                NUMBER(4)
ENAME                                VARCHAR2(20)
SAL                                  NUMBER(7,2)
```

Drop Emp2 table.

23.

```
SQL> drop table emp2;  
Table dropped.
```

24.

Flashback the first Emp2 table that contained data and check your table.

```
SQL> flashback table emp2 to before drop;  
Flashback complete.  
SQL> select * from emp2;  
no rows selected
```

25.

Flashback the recently dropped table by renaming to EMP2_1.

```
SQL> FLASHBACK TABLE emp2 TO BEFORE DROP RENAME TO emp2_1;  
Flashback complete.  
SQL> select * from emp2_1;  


| EMPNO | ENAME   | SAL  |
|-------|---------|------|
| 7839  | KING    | 5000 |
| 7788  | BLAKE   | 2850 |
| 7799  | JOHNSON | 2855 |


```

26.

Check your tables.

```
SQL> select table_name from user_tables;  


| TABLE_NAME  |
|-------------|
| REGIONS     |
| COUNTRIES   |
| LOCATIONS   |
| DEPARTMENTS |
| JOBS        |
| EMPLOYEES   |
| JOB_HISTORY |
| EMP         |
| EMP_TEST    |
| DEPT11      |
| EMP11       |
| EMP2        |
| EMP2_1      |

  
13 rows selected.
```